2.4. What proof exists that the photon has matter properties?

2.5. How can the neutrino be detected?

2.8. LITERATURE

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CHAPTER 3

Nuclei, Isotopes and Isotope Separation

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From a practical point of nuclear chemistry we accept the commonly used model that the nucleus is composed of only protons and neutrons. For the purposes of the scope of this book this definition is enough even if, as seen in Chapter 2, the world is far more complex than that. Although this is a simplification most observable patters of nuclear chemistry can be explained and the nomenclature can be kept simple. For example, the oxygen atom of mass number 16 has a nucleus which consists of 8 protons and 8 neutrons; since neutrons have no charge but are very similar to protons in mass, the net nuclear charge is +8. There are also 8 extranuclear electrons in the neutral atom of oxygen.

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